bridge. Measure the conductance. Take the given NaOH solution in a burette rinsed with the same solution. Rundown 0.2 cm^3 of NaOH solution into the beaker and shake well. Note down the conductance. Note down the conductance by adding 0.2 cm^3 of NaOH each time to the beaker. Conductance decreases in the beginning and finally starts increasing by large value. Take around 8-10 readings on the increasing side. Draw a graph of conductance against volume of NaOH added. The equivalence point in the graph corresponds to the neutralization of HCl. Calculate the normality and amount of HCl present in 1000 cm^3 of its solution.

Calculation

$$(NV)_{HCl} = (NV)_{NaOH}$$

$$N_{HCl} = \frac{(NV)_{NaOH}}{V_{HCl}}$$

Amount of HCl present in 1000 cm^3 of its solution = N_{HCl} x gram equivalent weight of HCl (36.5)

Model procedure /Flow chart

Pipette out 50cm^3 of acid into a 200 cm^3 beaker. Dip the conductivity cell in the acid such that two platinum electrodes are immersed completely. The cell is connected to conductivity bridge. Measure the conductance. Take the NaOH sol^n in burette rinsed with the same. Rundown 0.2cm^3 of NaOH sol^n each time to the beaker. Take down the readings ~~till~~ until you pipette out 4cm^3 of NaOH solution. At first the conductance decreases and then increases with large values. Draw the graph of conductance against volume of NaOH. The equivalence point corresponds to neutralisation of HCl.

Model calculation

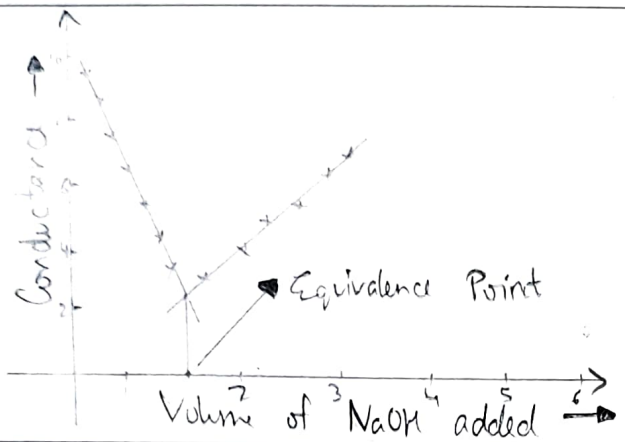
$$(NV)_{HCl} = (NV)_{NaOH}$$

$$N_{HCl} = \frac{(NV)_{NaOH}}{V_{HCl}}$$

Amount of HCl present in 1000cm³ of its solution =

N_{HCl} × gram equivalent weight of acid.

Model graph



Tabulation NaOH vs HCl

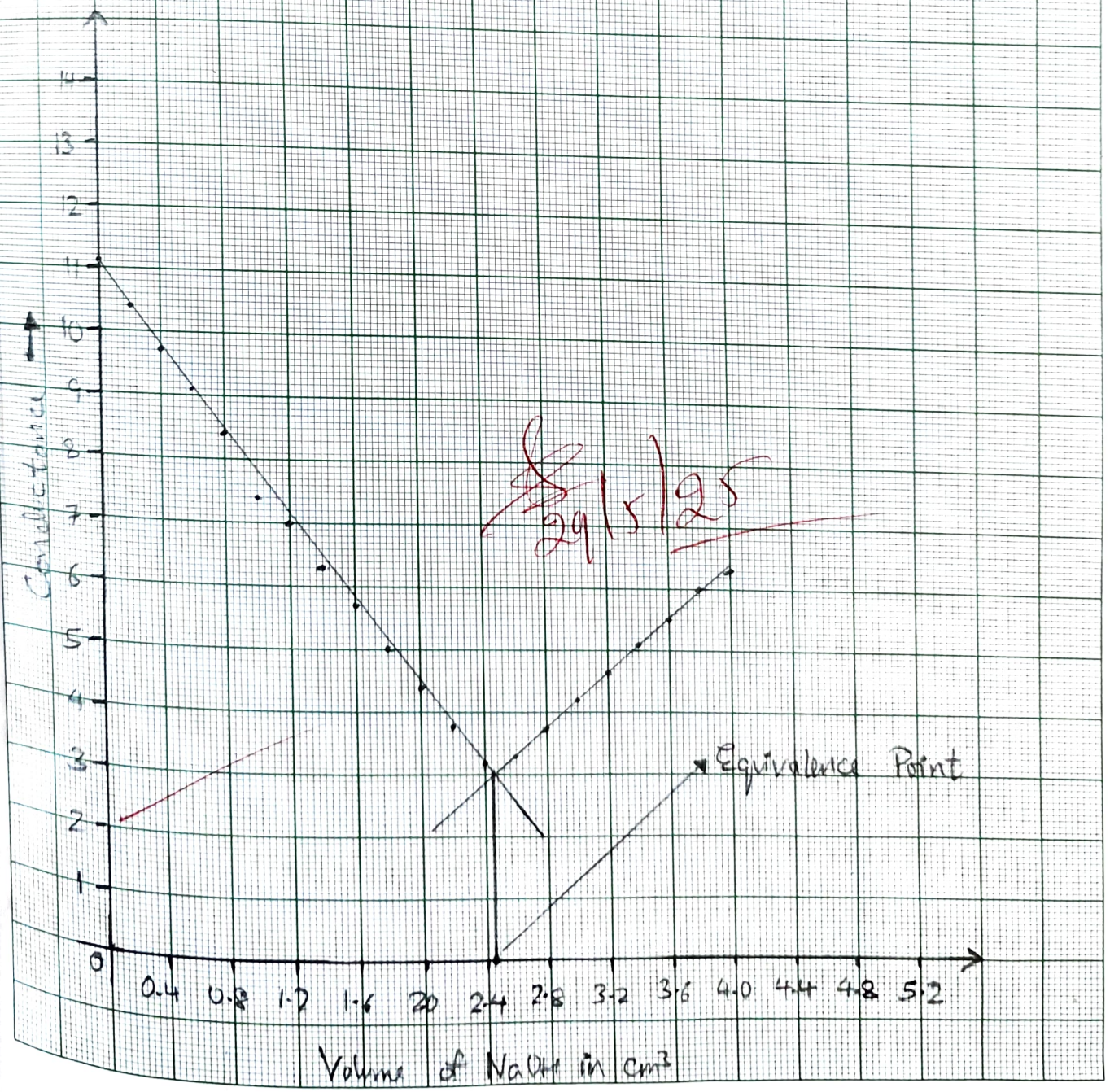
Volume of NaOH in cm³	Conductance in mS
0.0	11.1
0.2	10.4
0.4	9.7
0.6	9.1
0.8	8.4
1.0	7.4
1.2	7.0
1.4	6.3
1.6	5.7
1.8	5.0
2.0	4.4
2.2	3.7
2.4	3.2
2.6	3.3
2.8	3.7
3.0	4.2

Volume of NaOH in cm³	Conductance in mS
3.2	4.6
3.4	5.1
3.6	5.5
3.8	6.0
4.0	6.3
4.2	
4.4	
4.6	
4.8	
5.0	
5.2	
5.4	
5.6	
5.8	
6.0	

Scale:-

x-axis → 1 unit = 0.5 cm^3 NaOH

y-axis → 1 unit = 1 ms



29/5/25

Calculation

$$(NV)_{HCl} = (NV)_{NaOH}$$

$$N_{HCl} = \frac{(NV)_{NaOH}}{V_{HCl}} = \frac{0.049 \times 2.45}{50} = 2.401 \times 10^{-3} N$$

Amount of HCl present in $1000 cm^3$ of its sol^n :-

$= N_{HCl} \times$ gm equivalent of HCl

$= 2.401 \times 10^{-3} \times 36.5$

$= 0.0876$ g

Inference

The value of conductance tends to decrease during the acid-base titration until a certain point, after which it starts increasing by large amount. The equivalence point in the graph corresponds to neutralisation of HCl.

Relevance to society & environment

→ Conductometry can be used to examine the salinity of sea water and alkanity of fresh water.

→ This type of titration is used in food industries and food microbiologist to examine microorganisms.

→ Conductometry is used in medical diagnosis to analyse biological fluids.

Report

1. **Normality of HCl =** 2.401 x 10^{-3} **N**

2. **Amount of HCl present in 1000 cm³ of its solution =** 0.0876 **g**

Evaluation of experiment - 6		
Components	**Marks**	
	Max	**Obtained**
Model procedure, model graph & calculation	9 + 7	16
Equivalence point & execution	16 + 4	20
Inference & societal relevance	2 + 2	2+1
Total	40	38+1 = 39
Signature of teacher		29/5/25

Remove the liquid from the viscometer, clean well with acetone and dry it in oven. Cool the viscometer to room temperature. Similarly, measure the time of flow of water by taking the same volume. Calculate average time of flow of water.

Note down the room temperature. Knowing the density, viscosity coefficient of water and density of the given liquid, viscosity coefficient of the liquid can be calculated using the formula given below.

$$\eta_l = \frac{t_l d_l}{t_w d_w} \times \eta_w$$

(1 milli poise = $10^{-4} Nsm^{-2}$)

Model procedure /Flow chart

Take a clean and dry viscometer. The viscometer is fitted to a stand and is immersed in a waterbath such that its larger bulb is completely immersed in it fill the viscometer with the unknown liquid and suck the liquid slightly up to the upper mark of the small bulb of the viscometer. When the liquid crosses the upper mark, start the stopwatch and when it crosses the lower mark stop it. Note the time taken by it. Similiarly, repeat the process 3 times. After that the viscometer is washed with acetone and dried in an oven for 5 min. Then the similer process is done with water. for 3 time. Values for water is also taken and calculated for viscosity.

Model calculation

Room Temperature = T°C

Coefficient of Viscosity of Water = η_w

Coefficient of Viscosity of Liquid = η_L

Density of the Water = d_w

Density of the Liquid = d_L

$\therefore$ Viscosity of given liquid $(\eta_L) = \dfrac{\eta_w \times t_L \times d_L}{t_w \times d_w}$

where, t_L = avg. time of liquid

t_w = avg. time of water

Observation

Liquid	Time of flow		
	in min: sec	in seconds	Average in seconds
Test	1:52:53	112:53	t_l=112
	1:53:87	113:87	
	1:51:44	111:44	
Water	2:21:68	141:68	t_w=145
	2:26:84	146:84	
	2:27:97	147:97	
Temperature of water bath (°C)		28	
Density of water (g/cm^3)		0.9962	
Density of test liquid (g/cm^3)		0.8594	
Viscosity coefficient of water (milli poise)		8.360	

Calculation

Temperature = 28°C

Coefficient of viscosity of water ~~and liquid~~ = η_w = 8.360 millipoise

Avg. time of Liquid = 112 (t_L)

Avg. time of Water = 145 (t_w)

Density of Water = 0.9962 (d_w)

Density of Liquid = 0.8594 (d_L)

$\eta_L = \dfrac{\eta_w \times t_L \times d_L}{d_w \times t_w}$ = 5.5706 millipoise

Inference

From the above experiment we can infer that viscosity of given liquid can be calculated by using Ostwald's Viscometer and measuring the time taken by liquid to flow from upper mark to lower mark. By using the given data viscosity of given data comes out to be 5.57 millipoise Also, it is a temperature dependent characterstic.

Relevance to society & environment

Viscosity of a liquid is a factor used mainly in medical industry to determine the viscosity of cough syrup, etc. Gums should be highly viscous so that it has strong sticking action. It is also used in paint industry to make paints with proper viscosity.

Report

The coefficient of viscosity of the given liquid at 28.°C = 5.57 **X10^{-4} Nsm^{-2}**

Evaluation of experiment - 7		
Components	**Marks**	
	Max	**Obtained**
Model procedure, model graph & calculation	7 + 5	12
Expected value & execution	20 + 4	24
Inference & societal relevance	2 + 2	3
Total	40	39
Signature of teacher		29/8/25

Model procedure /Flow chart

2, 4, 6, 8 and 10 cm^3 of standard sodium solution into different 25cm^3 flasks from burette. Make all soln upto the mark using distilled water. and shake well. Switch on the instrument and light gas at burner. Adjust the air supply from compressor to 10lbs/sq inch. Place sodium filter in position. Now dip capillary tube in a cell containing distilled water. Adjust the flame photometer to zero by means of zero control knob. Feed the various sodium soln prepared, through the flame one by one including unknown soln. Note the readings. Plot the graph and using it find ~~soln~~ volm of unknown solution

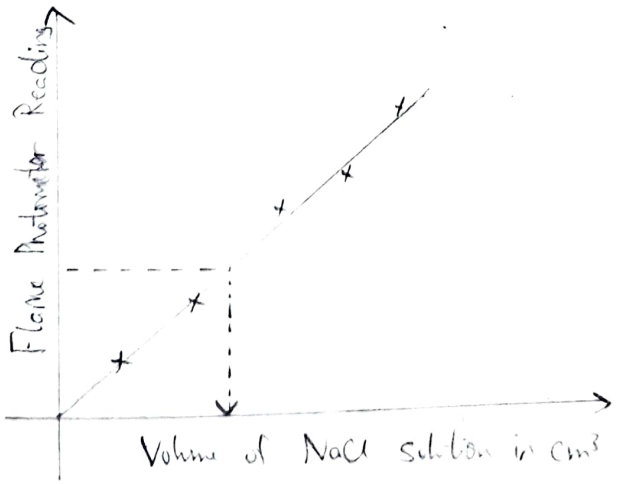
Model calculation

Amount of NaCl in given 1000cm^3 solution = 2.5g

58.5g of NaCl contain 23g of Na

∴ 1cm^3 of given stock soln contains = $\frac{w \times 23}{58.5 \times 100}$ = A g of Na

Model graph



Tabulation

Sl. No.	Vol. of NaCl in cm^3	Flame Photometer reading	Wt. of sodium in mg
1.	2	4.1	19.66
2.	4	5.9	39.32
3.	6	7.2	58.98
4.	8	8.2	78.64
5.	10	9.2	98.30
6.	Unknown	4.7	28.50

Calculation

Amount of NaCl in the given $100cm^3$ solution = 2.5g

58.5g of NaCl solution contains 23g of Na

$\therefore$ $1cm^3$ of given stock sol^n contains = $\frac{2.5 \times 23}{58.5 \times 100}$ = 0.00983g of Na

= 9.83mg of Na

$\therefore$ For:- $2cm^3 \Rightarrow 2 \times 9.83 = 19.66mg$

$4cm^3 \Rightarrow 4 \times 9.83 = 39.32mg$

$6cm^3 \Rightarrow 6 \times 9.83 = 58.98mg$

$8cm^3 \Rightarrow 8 \times 9.83 = 78.64mg$

$10cm^3 \Rightarrow 10 \times 9.83 = 98.30mg$

Unknown ($3.3cm^3$) $\Rightarrow$ 1.0 $\times$ 9.83 $\times$ 2.9

= 28.507mg

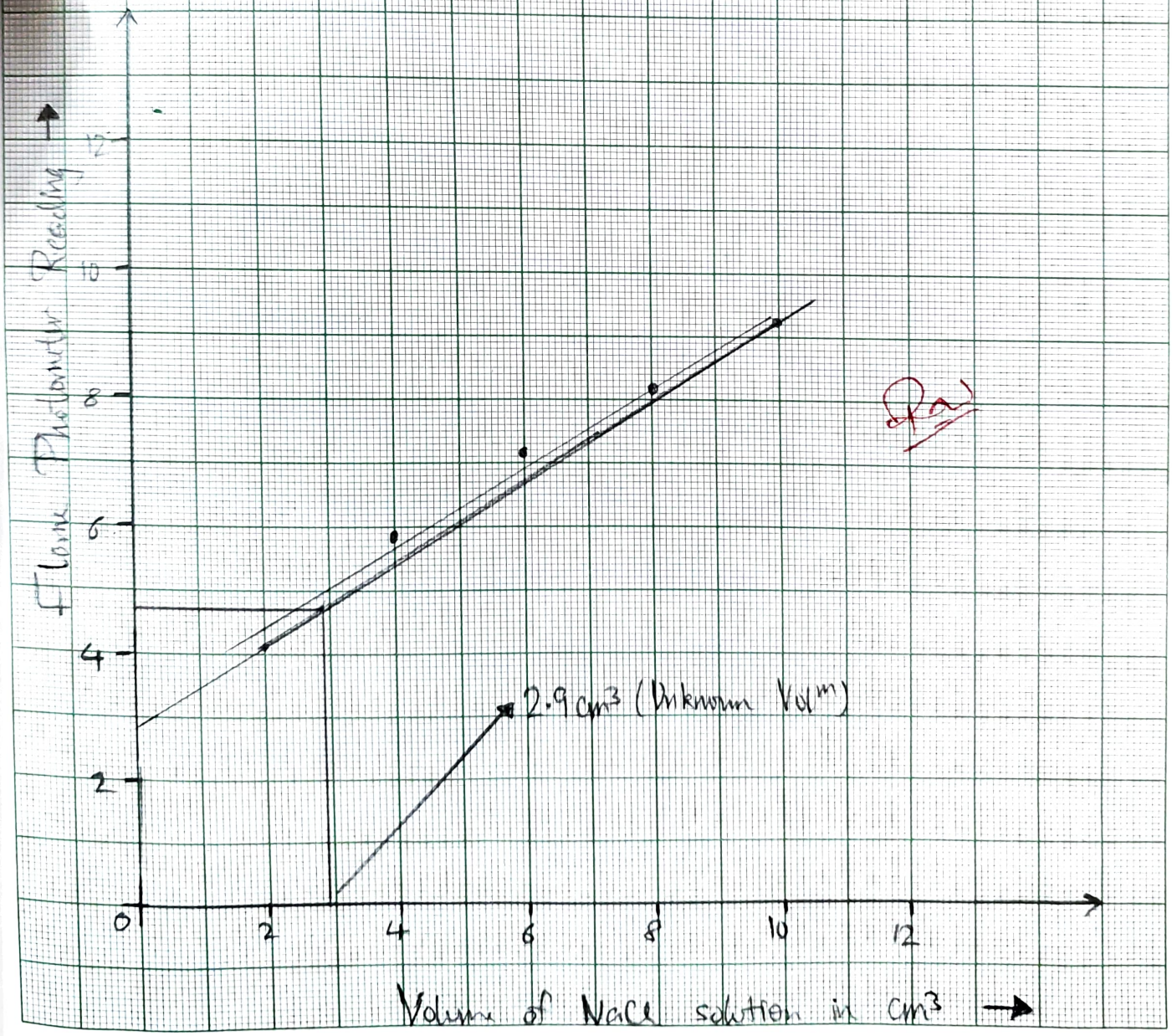
Inference

- Flame photometric estimation of sodium involves analysing the characteristic emission of sodium ions in flame
- It is an atomic emission technique used for detection of metals.
- It excites the electrons from ground state to higher state which when return to ground, emits radiation, which gets detected

Scale:-

$x \rightarrow 1\text{ unit} = 1\ cm^3\ NaCl$

$y \rightarrow 1\text{ unit} = 2$ unit reading



Relevance to society & environment

- This technique is commonly used for determining sodium level in diverse application.
- Some of these applications are -
  1) Food Analysis
  2) Environmental Monitoring
  3) Clinical Diagnostics

Report

1. **Volume of unknown solution =** 2.9 **cm³**
2. **Amount of sodium in the given unknown solution =** 28.507 **mg.**

Evaluation of experiment - 8		
Components	**Marks**	
	Max	**Obtained**
Model procedure, model graph & calculation	9 + 7	16
Expected volume & execution	16 + 4	20
Inference & societal relevance	2 + 2	3
Total	40	39
Signature of teacher		29/7/25